PAN-ATLANTIC UNIVERSITY

Installation Packages

Week 1 - Practical

Desmond Moru, PhD
Chukwudi Ofoma
SST/Computer Science

2026/2027 Session
Pan-Atlantic University

Today's Practical

Part 1

Meet the tools

GitHub, Git Bash, Sublime Text, Rust and PostgreSQL, and what each one is for.

Part 2

Set up GitHub

Create your account, create a repository and install Git Bash.

Part 3

Your first commit

Clone the repository, write a program, then add, commit and push it.

PART 1 OF 3

The Tools

What each software package does and why this course needs it.



The Software Packages



GitHub
Hosts your code online



Git Bash
Runs Git commands



Sublime Text
Where you write code



Rust
Our programming language



PostgreSQL
Stores data, used later on

Today we set up the first three. Rust and PostgreSQL join us in the coming weeks.

What is GitHub?

A home for your code

A website and cloud service where developers store and manage code, and track every change made to it.

Version control

Keeps the full history of a project, so a growing team can see, compare and undo changes safely.

Git underneath

Git is the distributed version-control system doing the work. Every developer has the full codebase and history, which makes branching and merging easy.



What is Sublime Text?

A source-code editor

A fast, cross-platform editor that natively supports many programming and markup languages.

Built for writing code

Syntax highlighting, quick file finding and a clean interface make it friendly for new programmers.

Where you will work

Every program you write in this course starts in Sublime Text.



What is Rust?

A systems programming language

Focused on three goals: safety, speed and concurrency.

Familiar syntax

Its syntax is similar to C++. Open source, originally developed at Mozilla Research.

Low-level control

You decide how memory is used: the stack for static allocation, the heap for dynamic allocation.



What is PostgreSQL?

An object-relational database

A powerful open-source database system with a strong reputation for reliability and performance.

Standards and extensibility

Also known as Postgres. Emphasizes SQL compliance and lets you extend it with your own types.

The data store behind apps

The primary database for many web, mobile, geospatial and analytics applications.



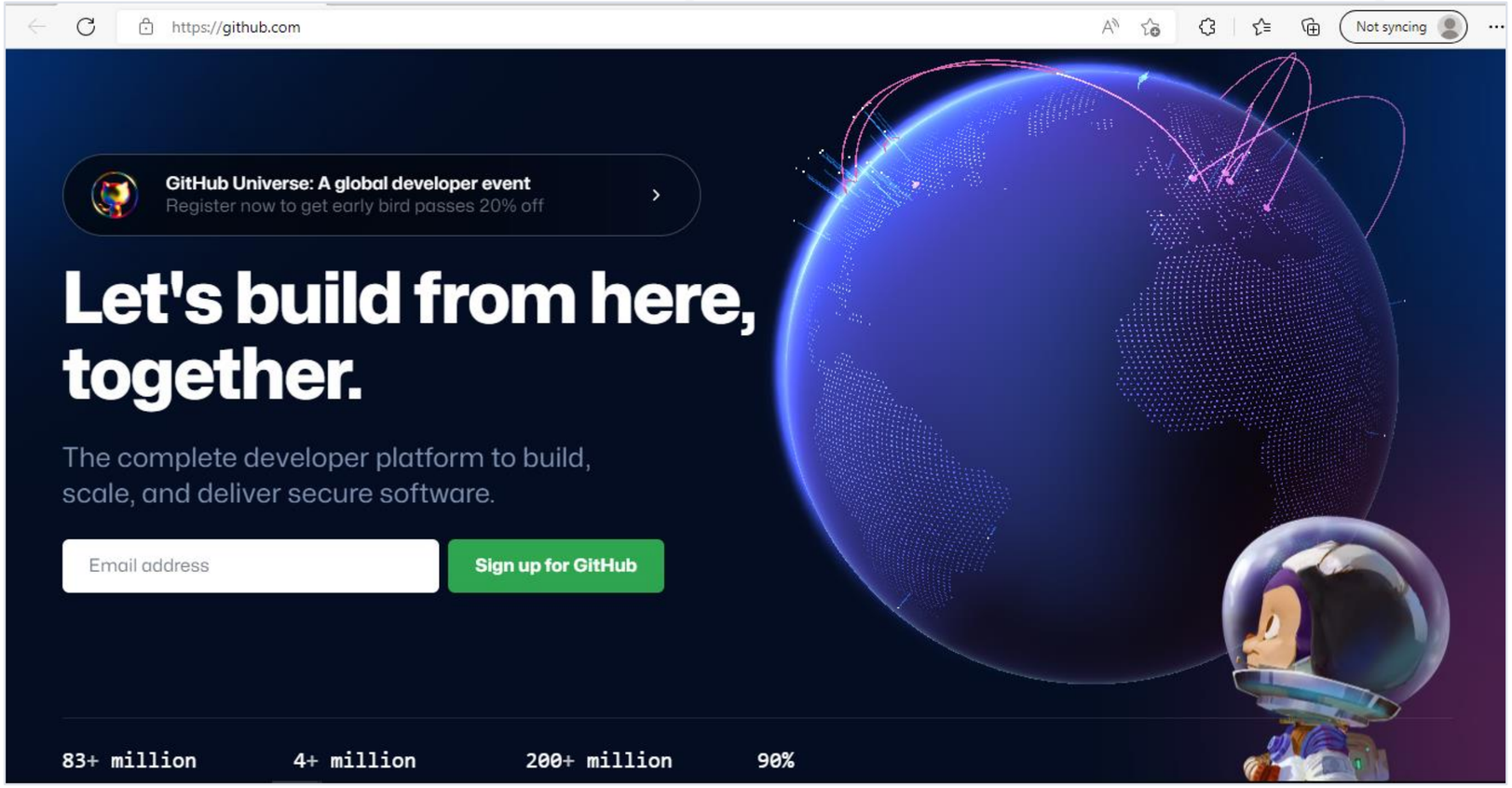
PART 2 OF 3

Set Up GitHub

Create your account and repository, then install Git Bash.



Set up a GitHub account

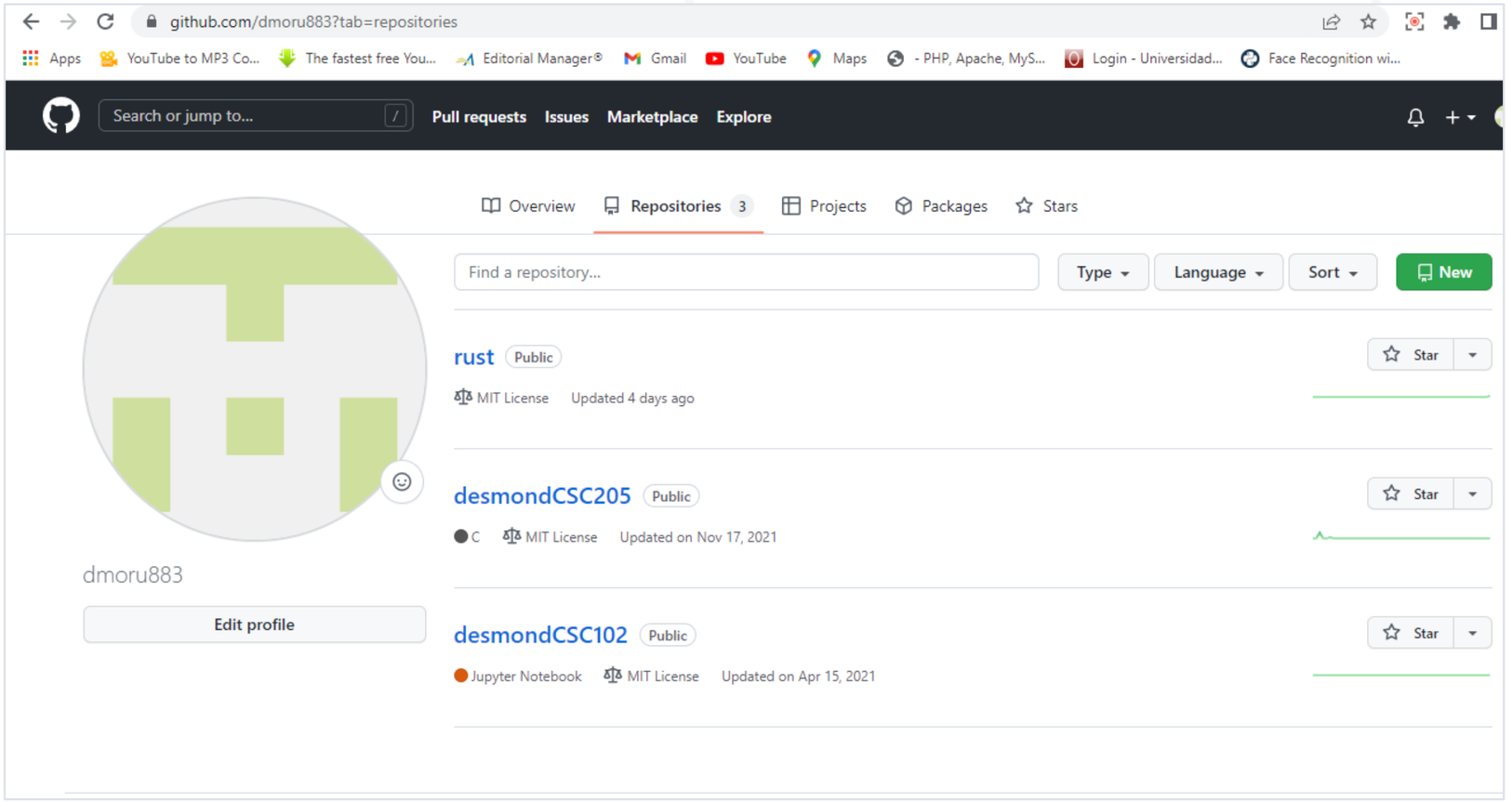


1. **Go to** `github.com/join`

2. **Type a username, your email address and a password**

3. **Choose “Sign up for GitHub” and follow the instructions**

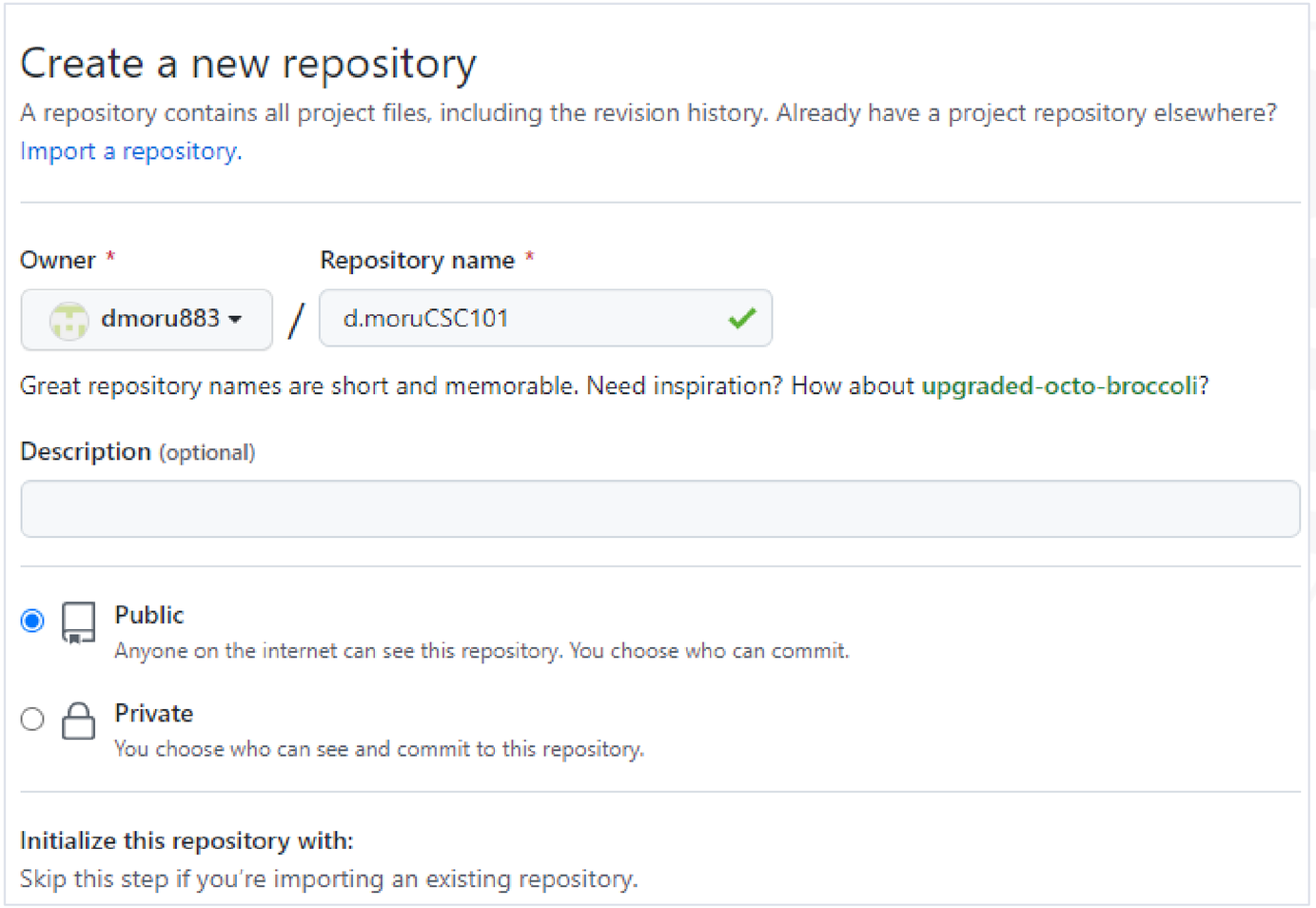
Create a repository



1. **Open your profile and go to Repositories**

2. **Click the green “New” button**

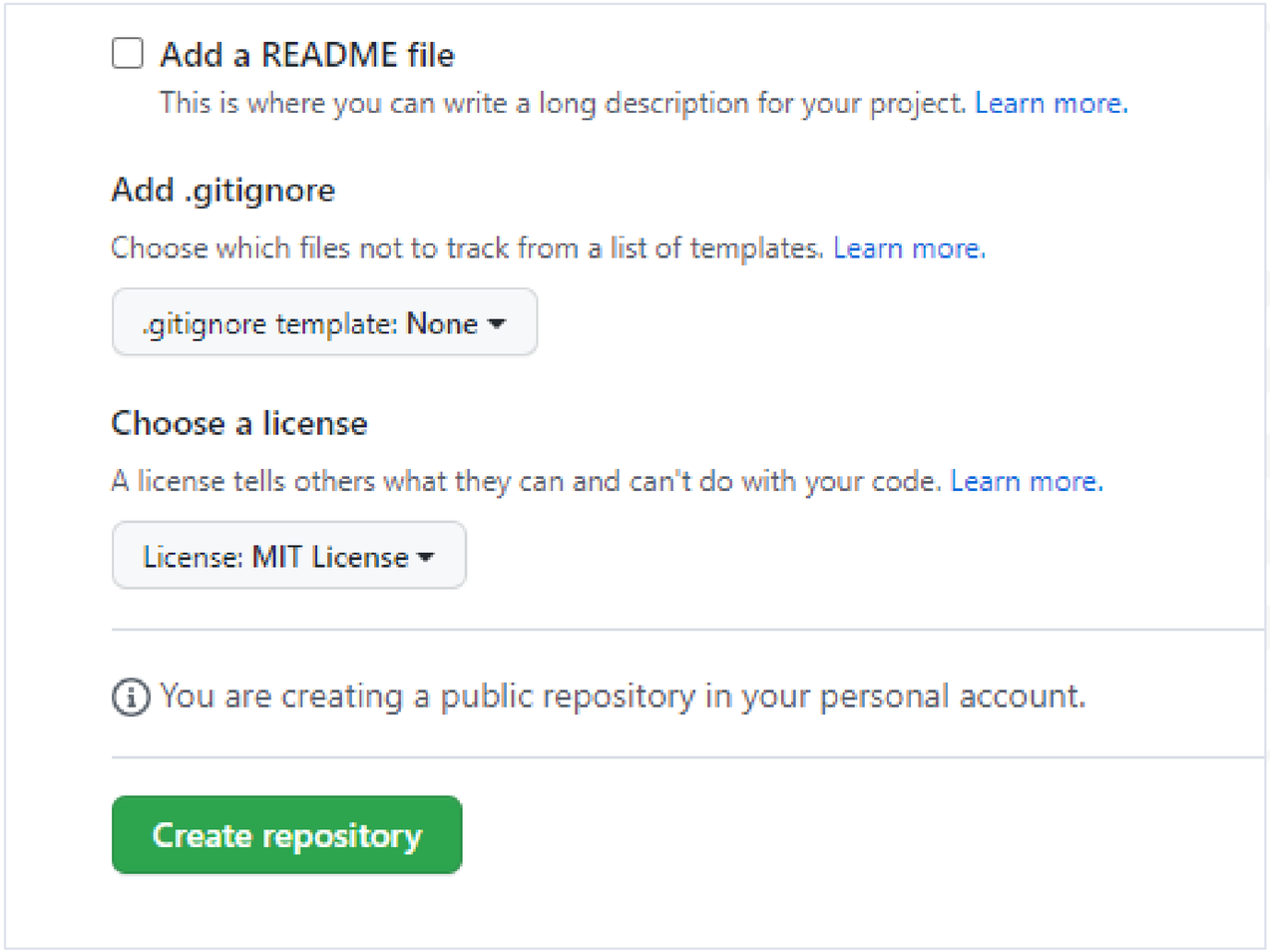
Name your repository



1. **Enter a repository name**
   Use yourname-COS101, e.g. d.moruCOS101

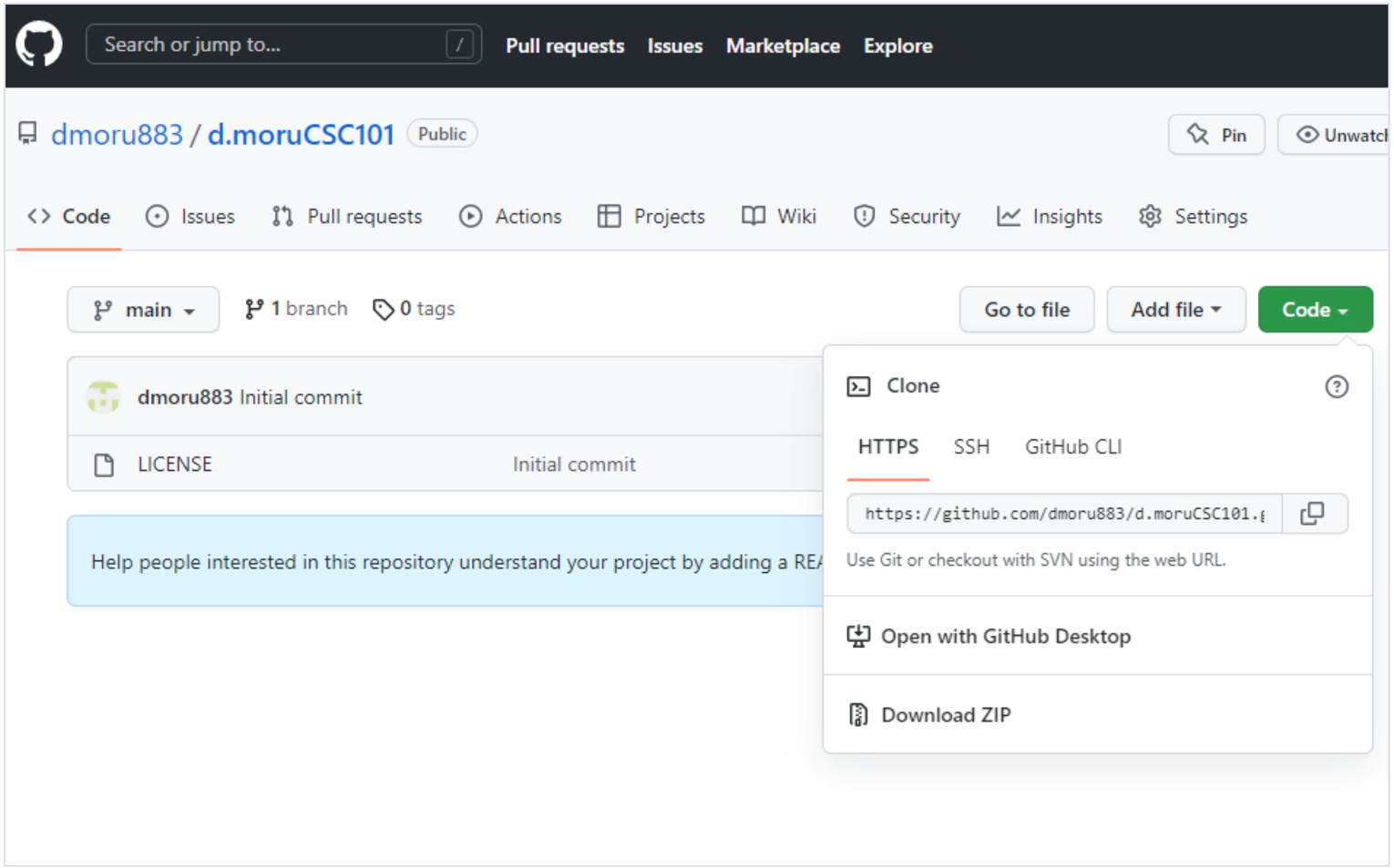
2. **Check "Public"**

Add a licence and create



1. **Choose the MIT Licence**
2. **Click “Create repository”**

Copy the repository URL



1. **Click the green "Code" button**

2. **Copy the HTTPS URL**
   You will need it to clone in Part 3.

Install Git Bash



1. **Go to**
   `git-scm.com/downloads`

2. **Download Git for Windows, Mac or Linux**

3. **Install it by following the instructions**
   Accept the default options.

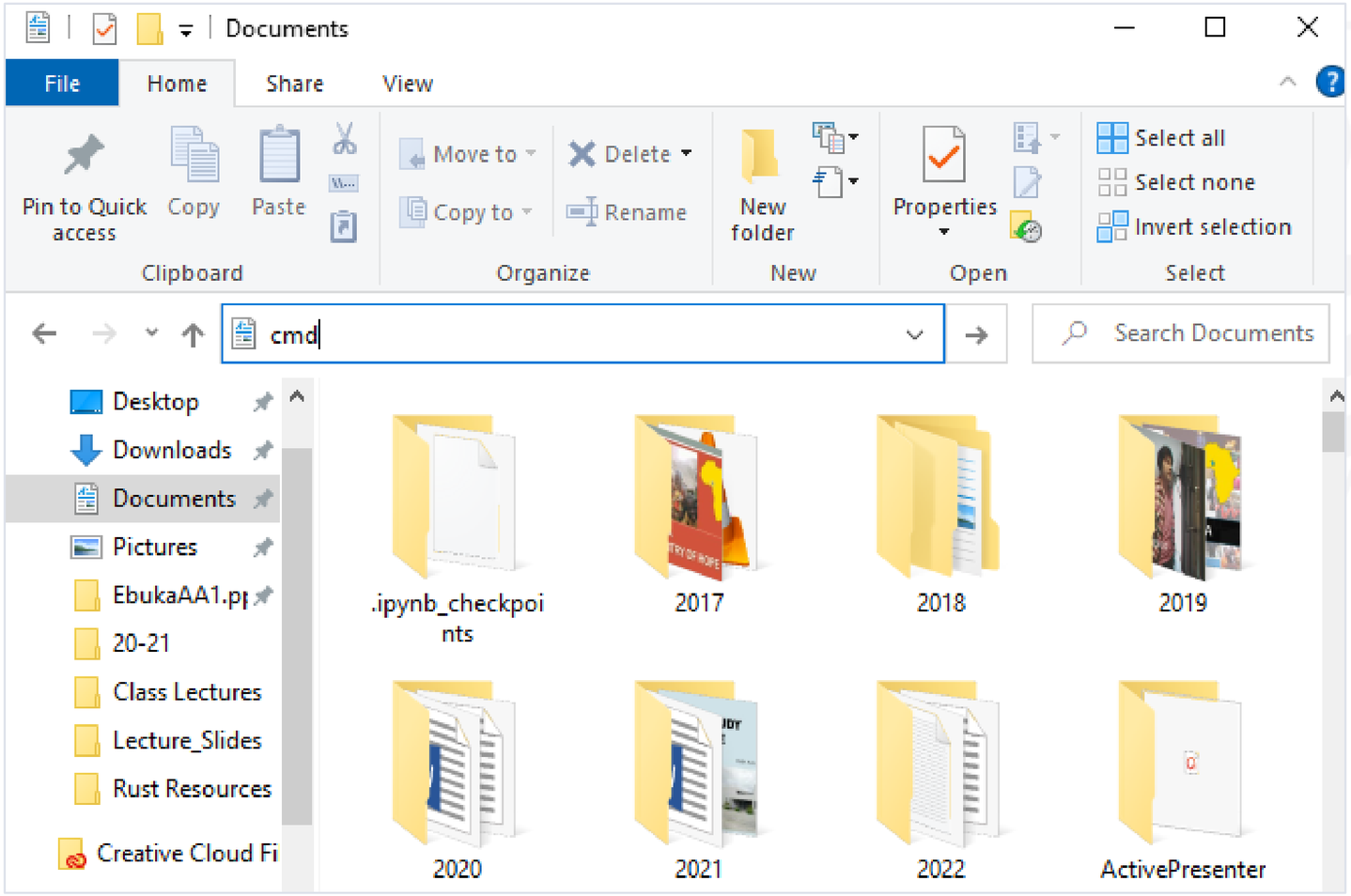
PART 3 OF 3

Your First Commit

From cloning the repository to pushing your first program.



Open a terminal in Documents

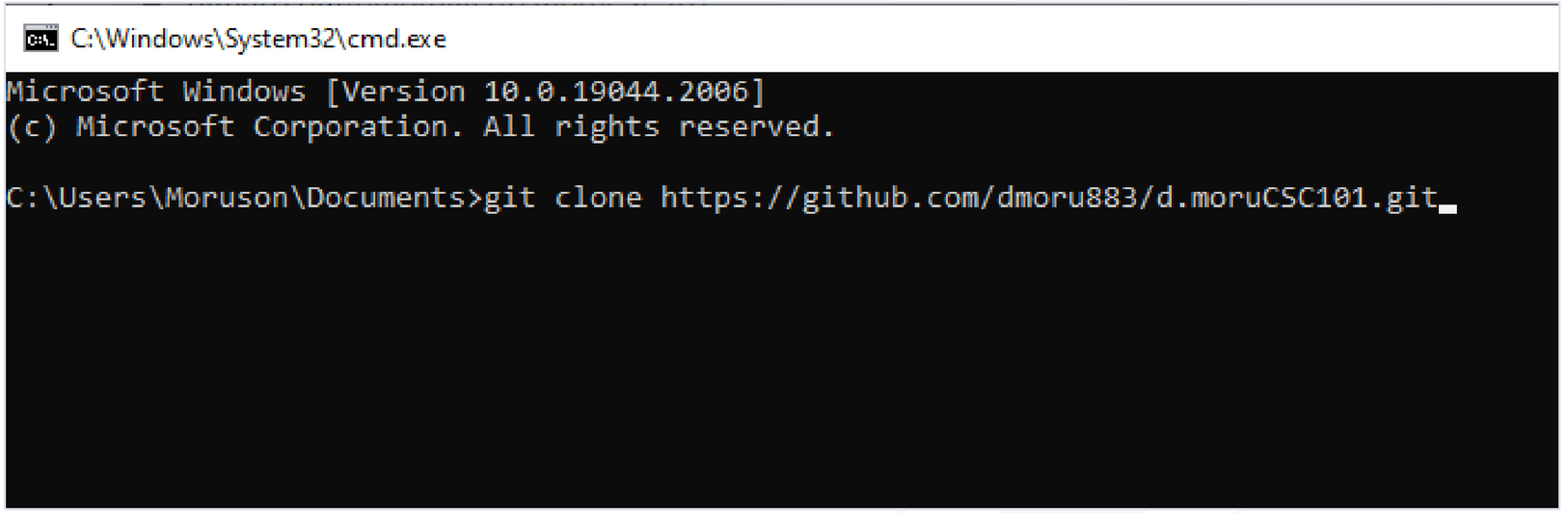


1. **Open your Documents folder**

2. **Type in the address bar**
   `cmd`

3. **Press Enter**
   A command prompt opens inside Documents.

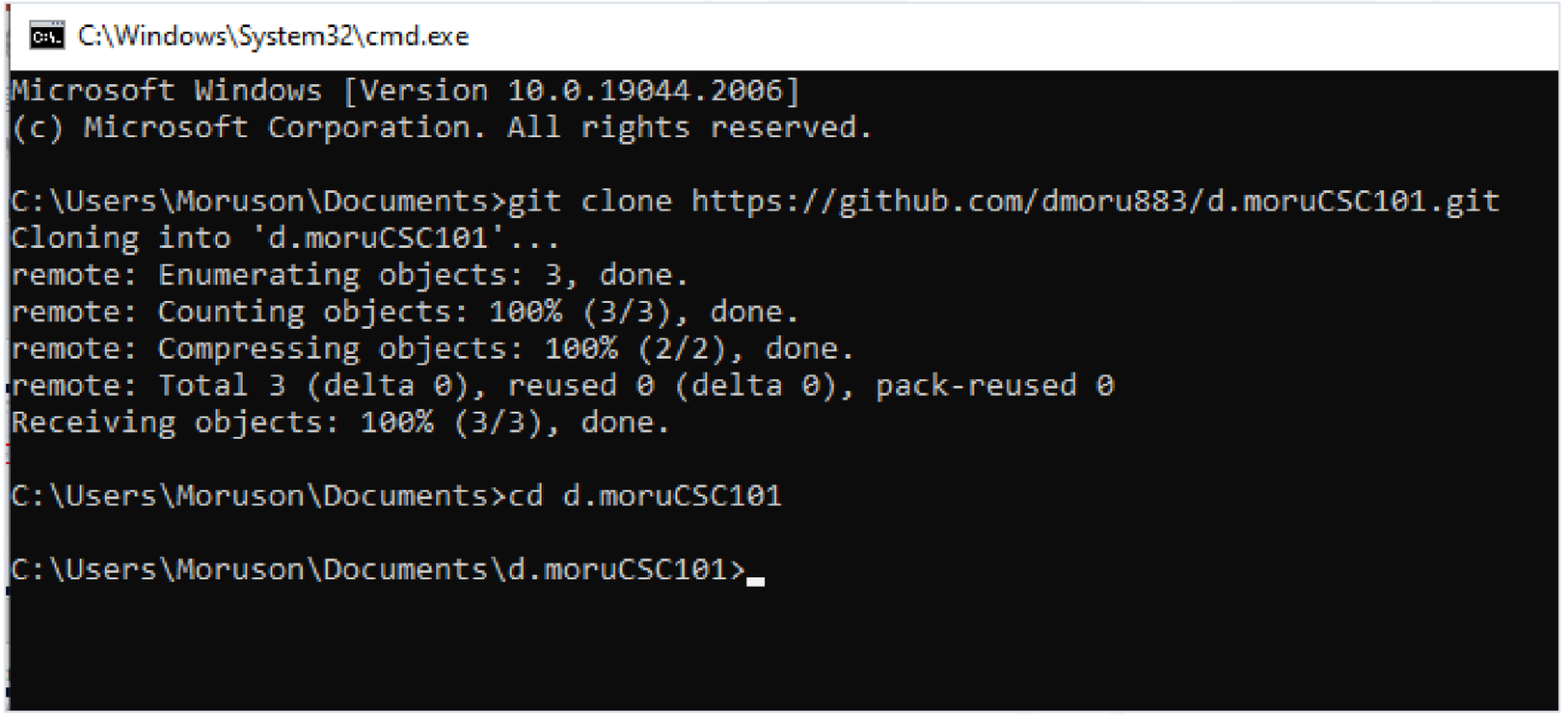
Clone your repository



1. **Type**
   `git clone <your-github-url>`

2. **Press Enter**
   Paste the URL you copied from GitHub.

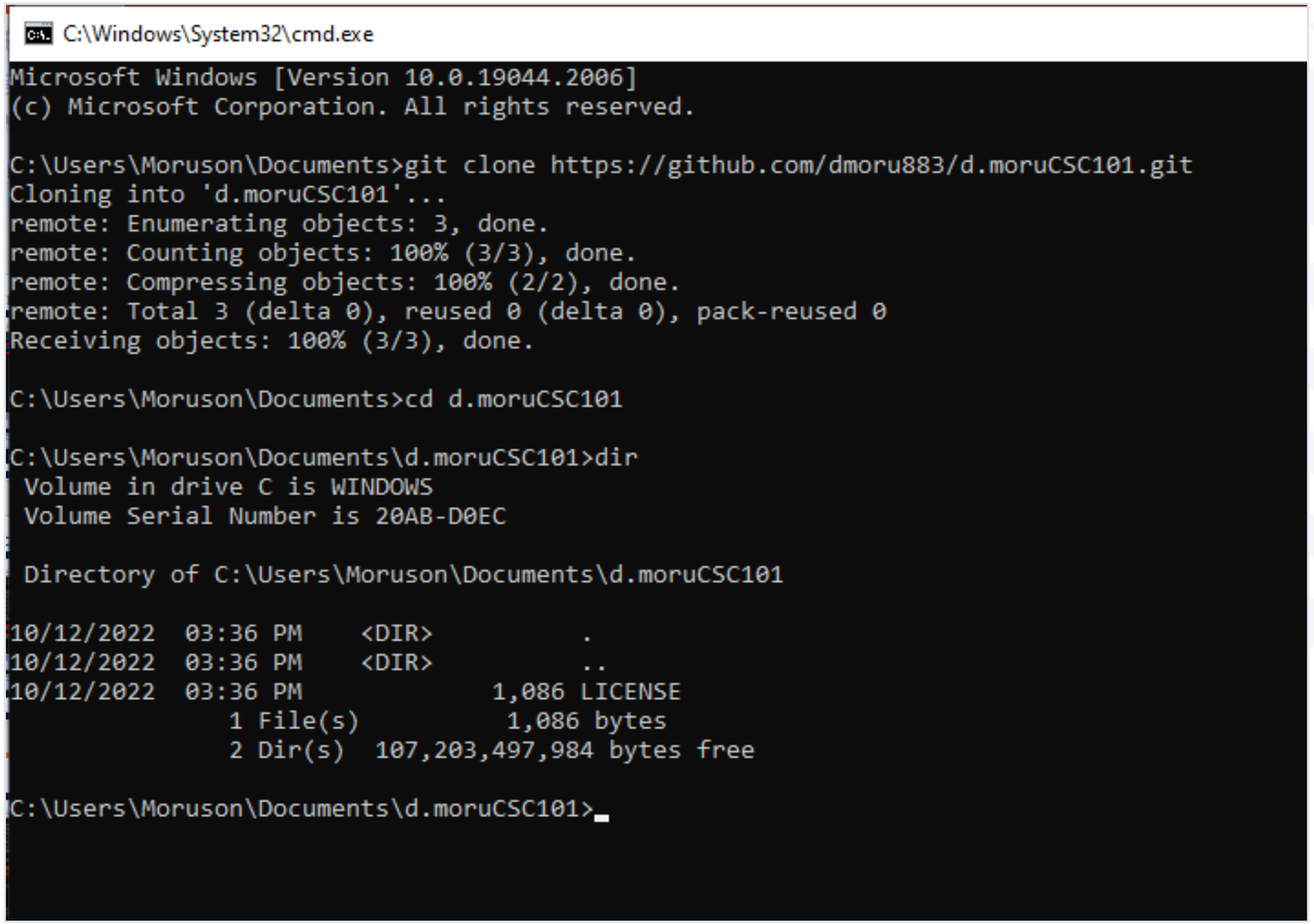
Enter the repository



1. **Type**
   `cd <your-repository-name>`

2. **Press Enter**

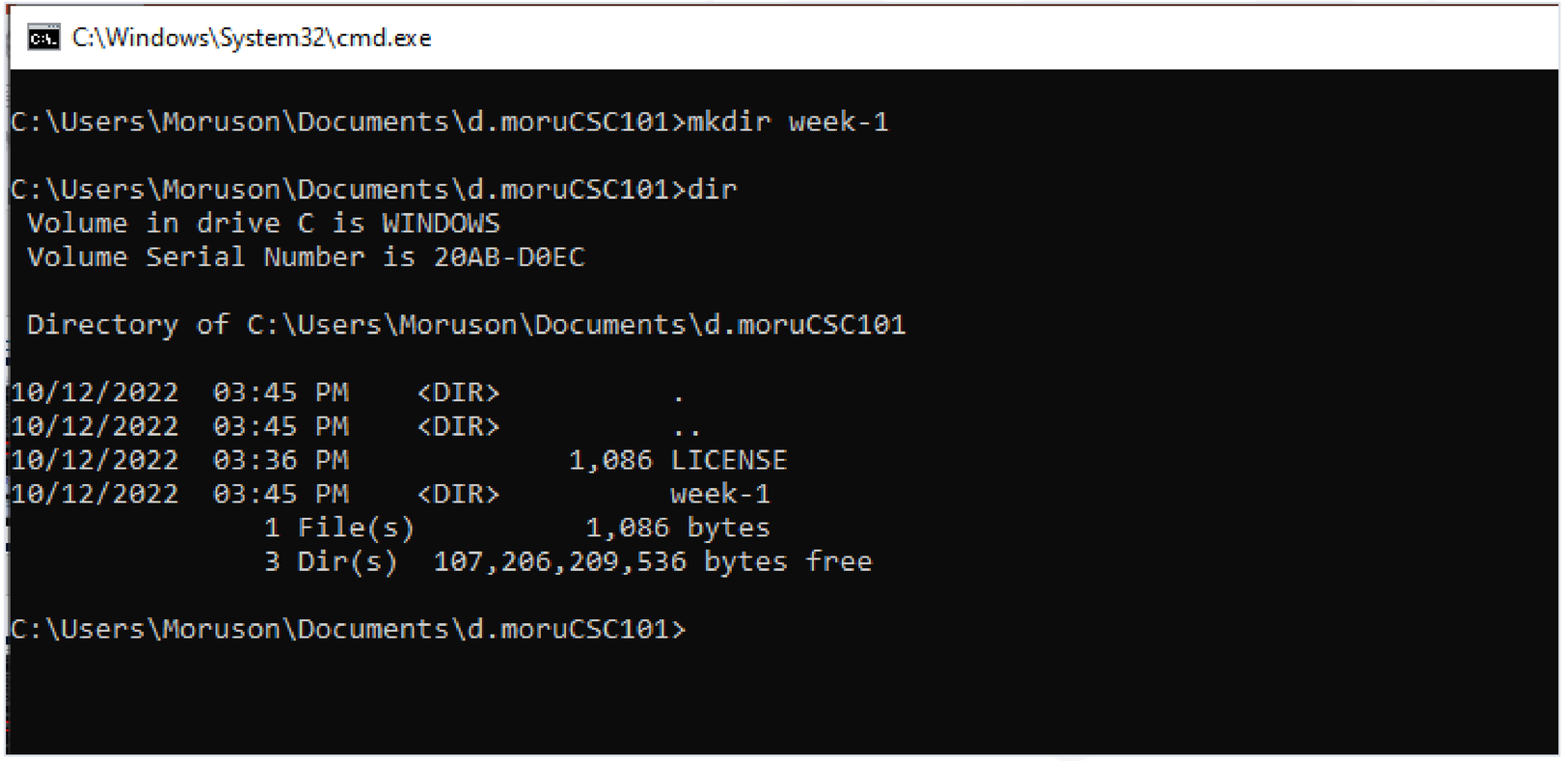
Check what is inside



1. **Type**
   `dir`

2. **Press Enter**
   dir lists every file and folder in the current directory.

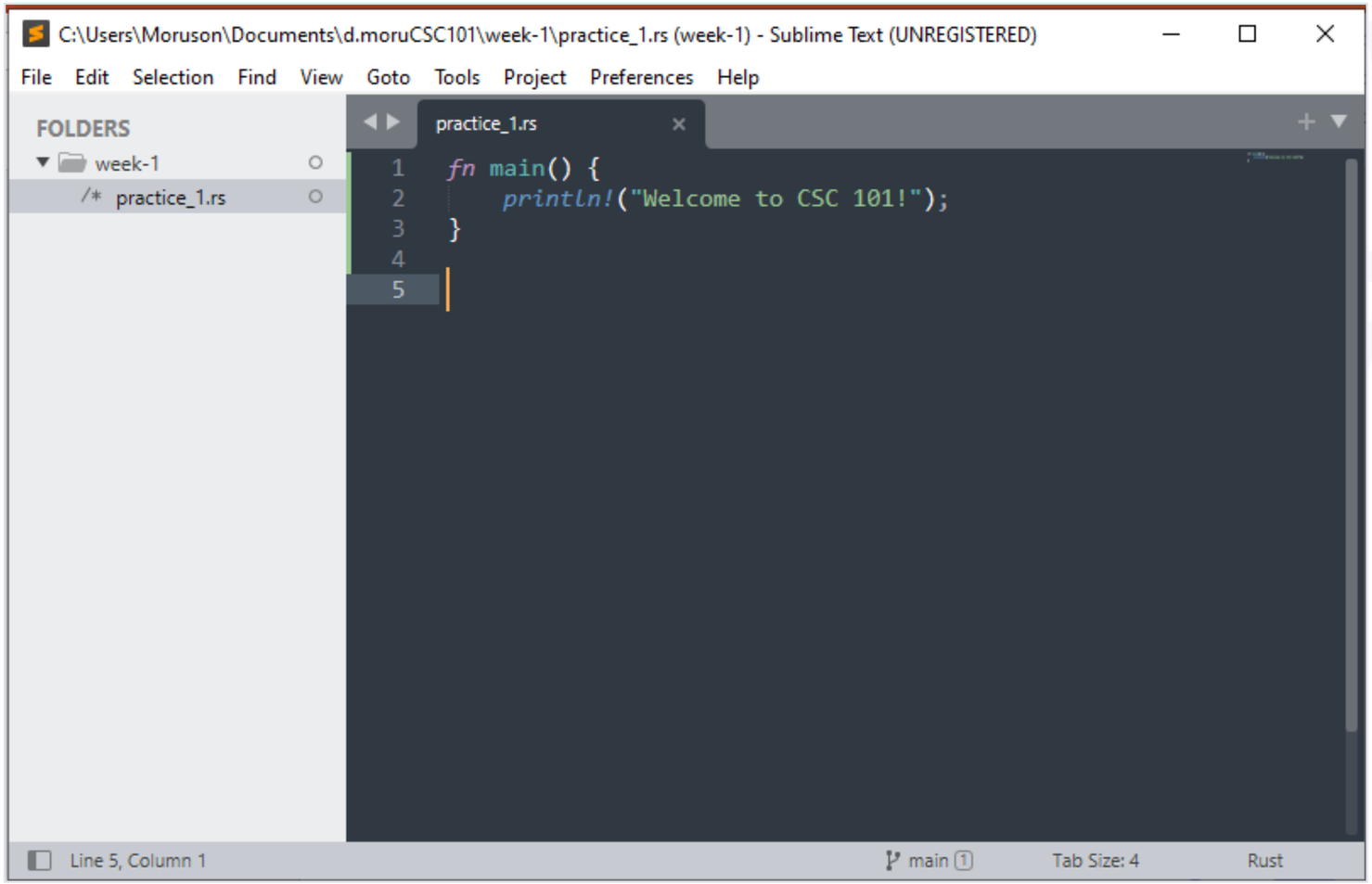
Create a week-1 folder



1. **Type**
   `mkdir week-1`

2. **Press Enter**

Write your first program

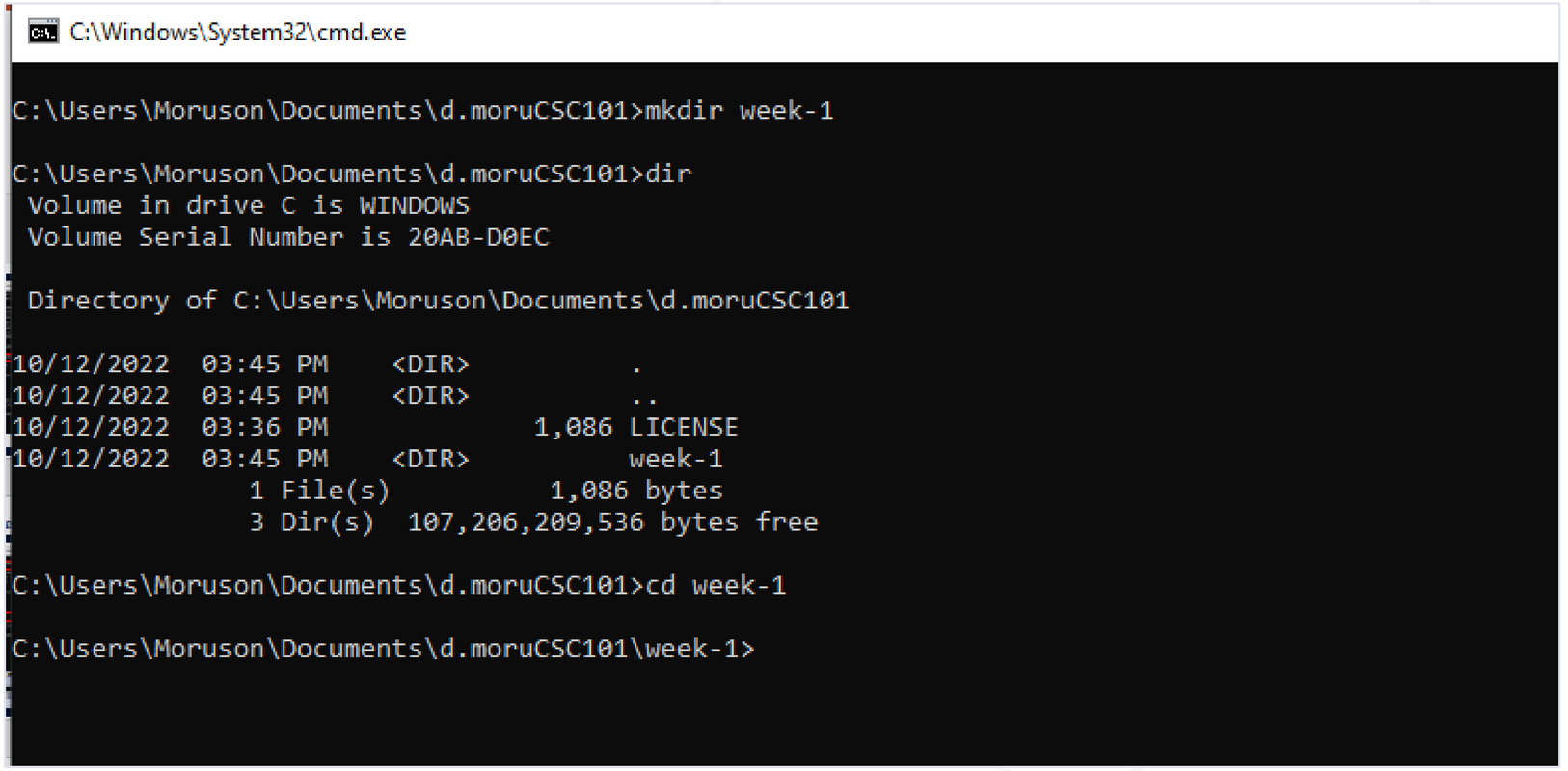


1. **Open Sublime Text and type the code**

2. **Save it as** `practice_1.rs`

3. **Save inside the week-1 folder of your cloned repository**

Move into week-1

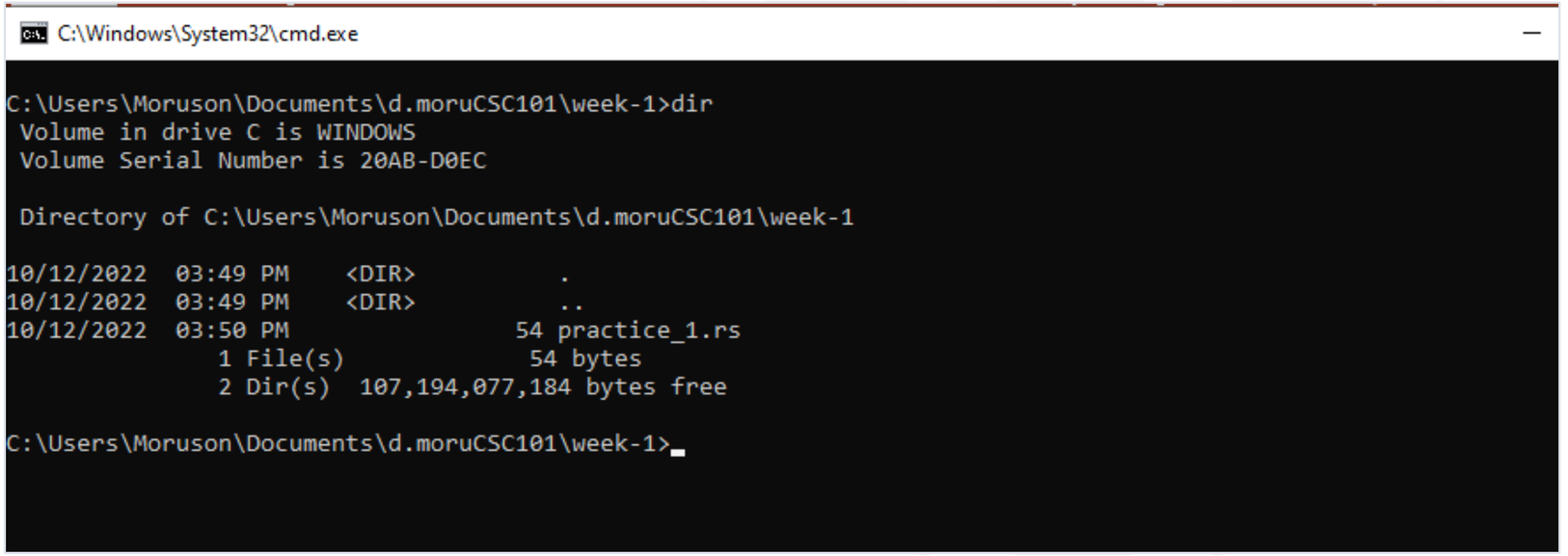


1. **Type**
   `cd week-1`

2. **Press Enter**

Tip: cls clears the screen whenever the terminal gets crowded.

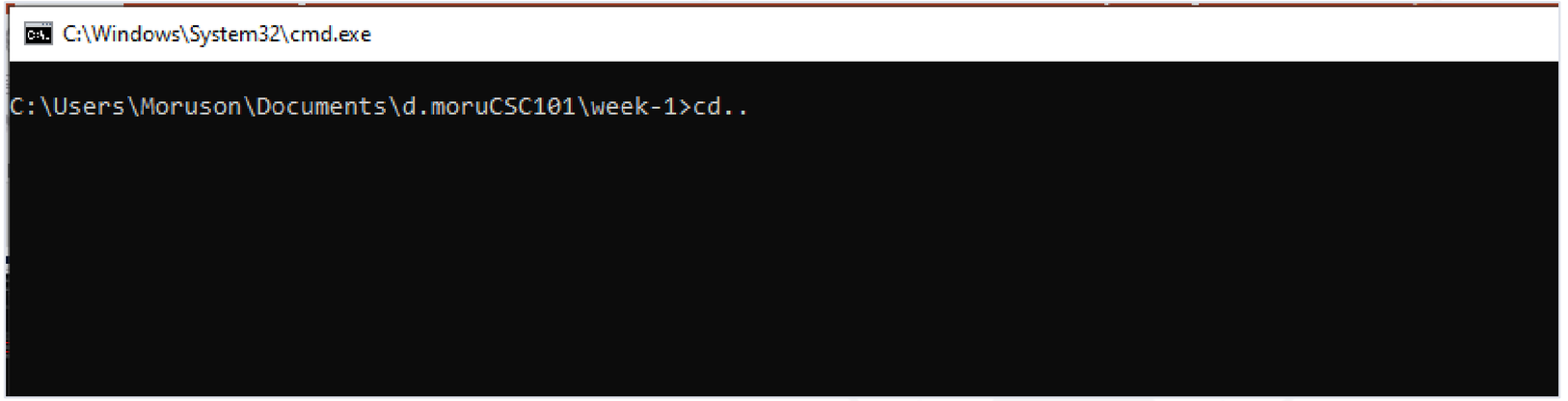
Confirm your file is there



1. **Type**
   `dir`

2. **Press Enter**
   You should see practice_1.rs listed.

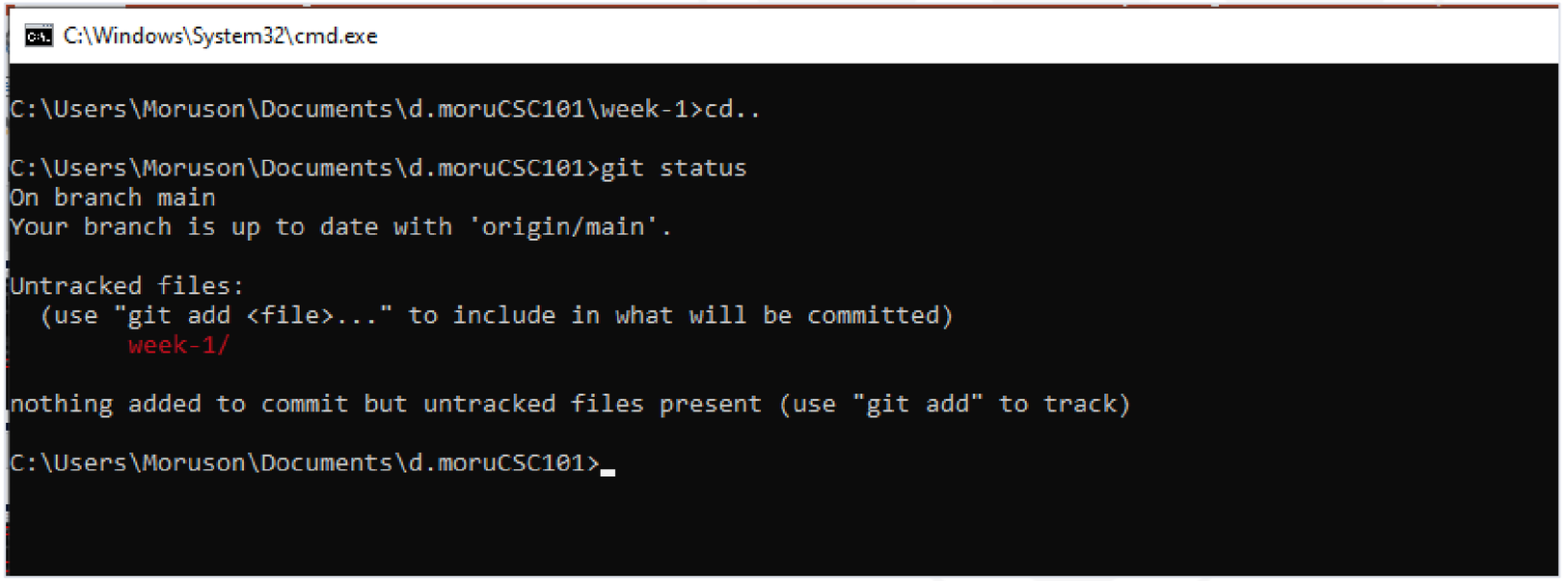
Back to the repository root



1. **Type**
   `cd ..`

2. **Press Enter**
   Git commands run from the repository root.

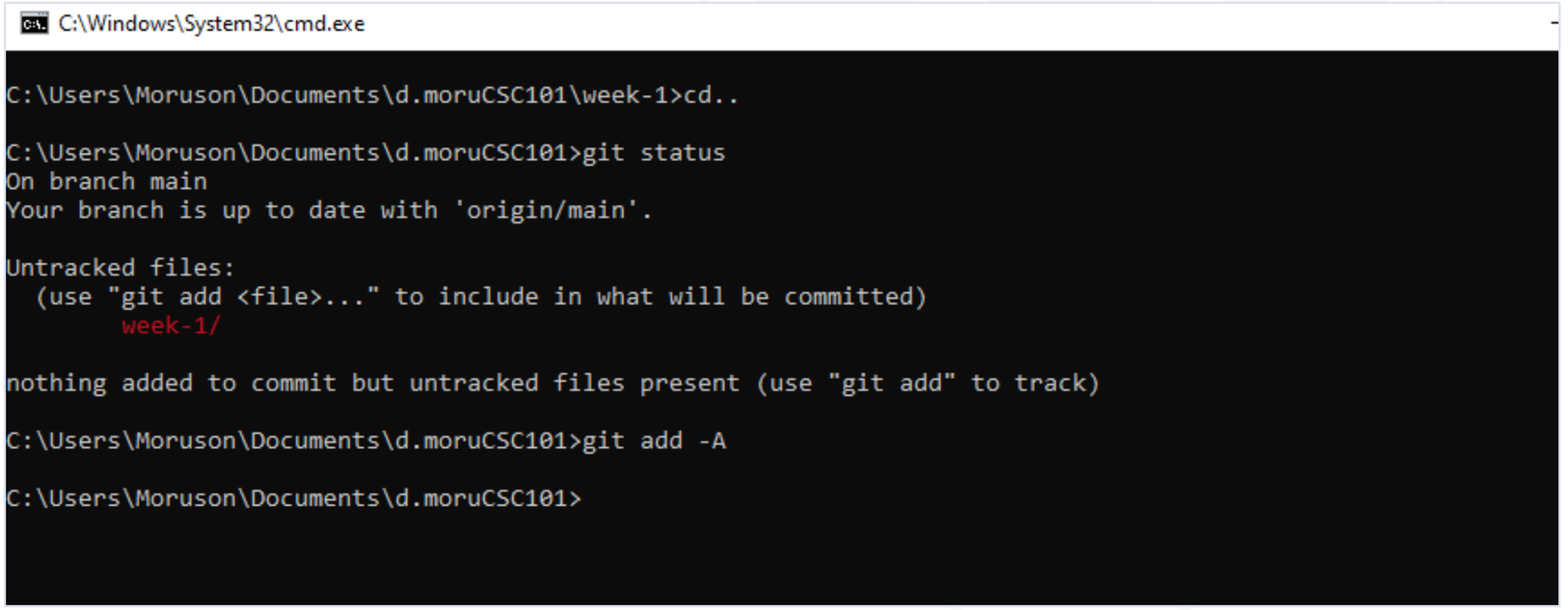
Check the status



1. **Type**
   `git status`

2. **Press Enter**
   Git shows week-1 as untracked since it is new.

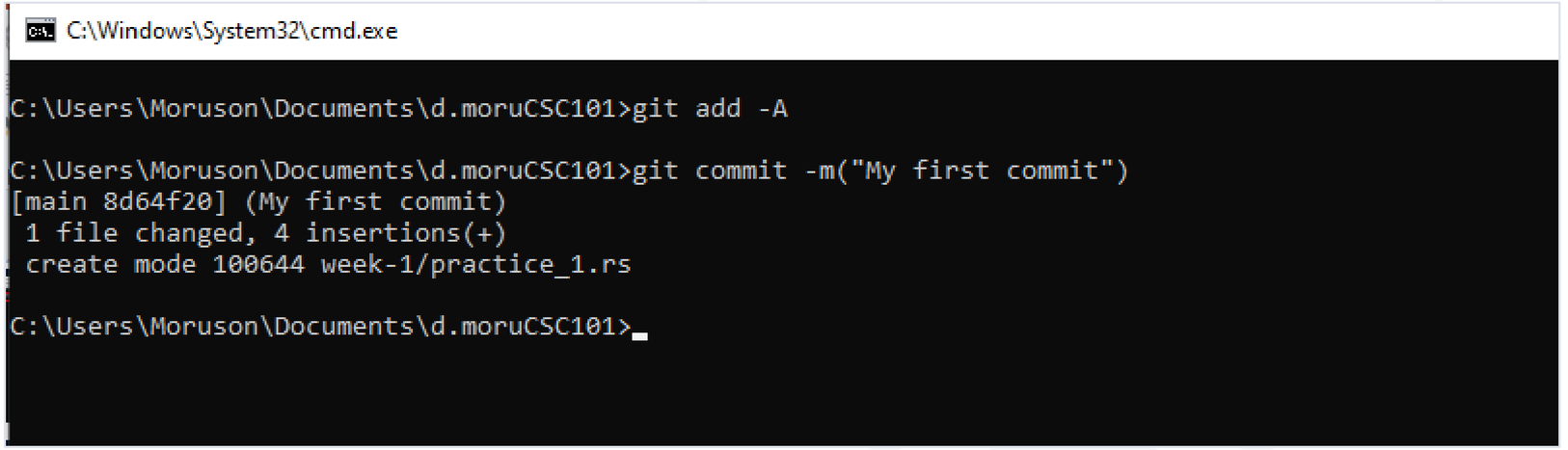
Stage your files



1. **Type**
   `git add –A`
   Or
   `git add .`

2. **Press Enter**
   "**-A**" or "**.**" stages every new and changed file.

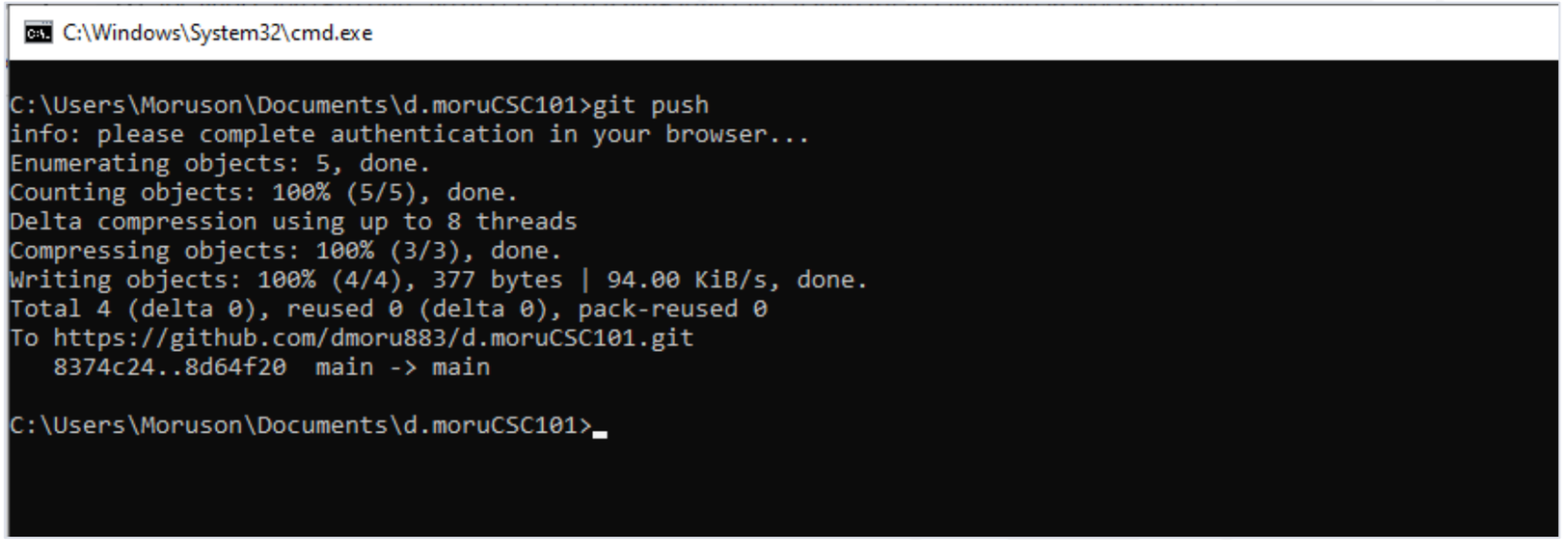
Commit your work



1. **Type**
   `git commit -m "My first commit"`

2. **Press Enter**
   The message describes what this snapshot contains.

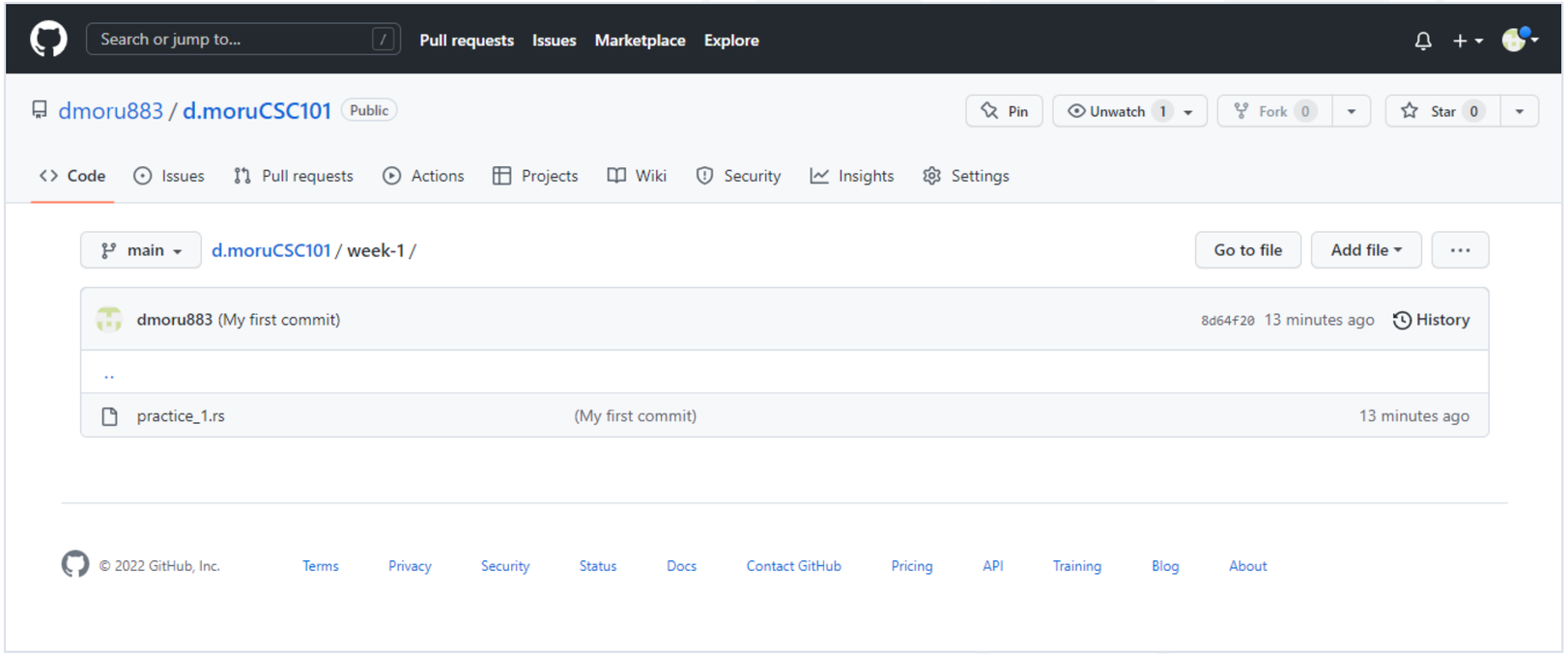
Push to GitHub



1. **Type**
   `git push`

2. **Press Enter**
   Sign in through the browser if Git asks you to.

Check it on GitHub



1. **Refresh your repository page in the browser**

2. **You should see week-1 with practice_1.rs inside**

The Whole Workflow

01 · Get the code

```
git clone <url>
cd <repository>
```

02 · Make changes

```
mkdir week-1
```

write code in Sublime

save as practice_1.rs

03 · Record them

```
git status
git add -A
git commit -m "..."
```

04 · Publish

```
git push
```

refresh GitHub

Same four moves every week. By Week 3 this is muscle memory.

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Questions?

Homework: push `practice_1.rs` before the next class.